

BirdsEye: Enabling Rapid Supervised Image Dataset Creation for Geospatial-AI

Morgan Masters, Adam Korycki, T. Luca Altaffer, Nick Kuipers, Nick Bender, Colleen Josephson, Steve McGuire

Introduction

High-quality labeled datasets are essential for modern machine learning, but collecting annotated imagery in outdoor domains such as agriculture, ecology, and infrastructure monitoring is labor-intensive and costly. Existing large-scale datasets and annotation tools are often biased toward urban or well-funded applications, leaving agricultural and ecological practitioners underserved. UAVs offer a scalable solution for capturing high-resolution imagery, yet the bottleneck remains in creating annotated datasets suitable for training deep learning models.

This work introduces **BirdsEye**, a UAV-based geospatial annotation pipeline that enables rapid, semi-automated dataset creation through tight integration of RTK-GPS, onboard cameras, and human-in-the-loop annotation. BirdsEye replaces time-consuming manual image labeling with structured ground-based geotagging, dramatically increasing annotation throughput while maintaining high spatial accuracy.

Methods

BirdsEye combines three key components:

1. **Geoannotation System:** Handheld RTK-enabled devices allow field experts to place centimeter-level accurate geotags of target features (e.g., pest burrows, diseased plants).
2. **UAV Data Capture:** An RTK-equipped UAV autonomously surveys the annotated field, capturing high-resolution imagery at 2–4 mm ground sampling distance (GSD).
3. **Annotation Pipeline:** Using precise projective calibration, BirdsEye maps ground geotags onto UAV imagery, generating pixel-level training annotations. The system supports batch post-processing, dataset export, and AI model training.

A case study on **pest burrow detection** demonstrates the pipeline. Images were tiled into 224×224 patches and labeled using Gaussian heatmaps around projected geotag locations. An EfficientNetV2-based CNN was trained and evaluated on a geographically distinct test site.

Results

- BirdsEye generated **55,600 annotations across 12,500 images in ~12 hours of labor** - less than a single workday for two field workers.
- Compared to GUI-based labeling, BirdsEye improved annotation throughput by **300–500%** while requiring **7× fewer workers**. Per-worker efficiency improved by **22–38×**.
- Projection ground sample distance was validated at **2.2 mm GSD (10 m altitude)** and **4.4 mm GSD (20 m altitude)**, with **sub-30 pixel**, or **sub-10 cm**, projection accuracy under normal flight conditions.
- The burrow detection CNN achieved **83.7% recall** on unseen farmland. While precision was modest (35.1%), many false positives were visually confirmed as missed positives by humans, revealing annotation limitations rather than model error.

Discussion

BirdsEye demonstrates that high-fidelity, field-ready datasets can be rapidly created using semi-automated geospatial annotation. The approach enables scalable model training in outdoor robotics without dependence on costly manual labeling or synthetic datasets. Limitations include degraded performance during UAV yaw maneuvers and challenges in exhaustive human annotation.

Beyond technical performance, ethical considerations are critical: automation should augment, not replace, expert knowledge. Transparency, community participation, and human-in-the-loop validation must remain central to sustainable deployment.

Conclusion

BirdsEye provides a reproducible, open-source framework for efficient dataset creation in geospatial AI. By coupling UAV autonomy, RTK georeferencing, and structured expert annotation, it offers a **democratized pathway for agricultural automation and ecological monitoring**. Future work will focus on multiclass annotation, improved flight planning, and semi-supervised learning protocols.